

Bahaeddine Krifa

Full-Stack Software Engineer

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PROFESSIONAL SUMMARY

Final-year software engineering student at the Polytechnic School of Sousse, seeking a 6-month end-of-studies internship (PFE). Full-stack developer working with Angular, React, Spring Boot and MySQL/MongoDB, with hands-on experience integrating third-party APIs (Meta Graph API, Google Workspace) and building real-time features. IEEE Student Branch Chair and Python instructor with proven leadership, teamwork and event-coordination skills. Fast learner, available for new opportunities.

EDUCATION

Engineering Degree in Software Engineering

Polytechnic School of Sousse, Tunisia | Sep 2025 – Present (final year, expected graduation 2027)

Engineering Degree in Applied Computer Science (Year 1)

National Engineering School of Sousse (ENISO), Tunisia | Sep 2024 – Jun 2025

Bachelor's Degree in Information Technology (Web Development & Multimedia)

Higher Institute of Technological Studies of Sousse (ISET Sousse), Tunisia | Sep 2021 – Jun 2024

Baccalaureate in Technical Sciences

Sousse High School, Tunisia | Sep 2017 – Jun 2021

WORK EXPERIENCE

Python Programming Instructor — Opus Lab

Sousse, Tunisia | Jun 2025 – Oct 2026

- Teach Python and algorithmic problem-solving to high school students through structured, hands-on sessions.
- Cover core concepts (variables, control structures, loops, functions) and guide students through step-by-step algorithm design.
- Adapt content and pace to each group so students build small real-world programs using core Python only.

Full-Stack Developer (End-of-Studies Internship) — Medical Innovation Technology Company

Tunis, Tunisia | Jan 2024 – Jun 2024

"MITIA": candidate-tracking web platform centralizing Facebook, Messenger and Gmail communication for recruiters.

- Designed and developed a full-stack web platform (Angular, Spring Boot, MySQL) that unifies Facebook, Messenger and Gmail interactions in a single interface.
- Integrated the Facebook Graph API to publish offers, track post engagement (likes, comments, shares), reply to comments and chat with candidates via Messenger.
- Built the Gmail integration with Spring Boot Mail (SMTP/IMAP) to import emails by date, detect new candidates and send replies from the app.
- Connected the Google Sheets API (GCP service account) to import Google Forms candidate data, and stored messages, contacts and posts in a centralized MySQL database.
- Implemented candidate status tracking (from registration to acceptance or rejection), filterable candidate lists and role-based user management (Agent, Manager, Administrator).
- Structured the app with a layered MVC 3-tier backend (Controller/Service/Repository, JPA) and an MVVM Angular frontend; modeled it in UML and worked in Agile iterations.

Technologies: Spring Boot, Angular, TypeScript, MySQL, JPA/Spring Data, REST APIs, Facebook Graph API, Gmail (SMTP/IMAP), Google Sheets API, Google Cloud Platform, Maven, Postman, MVC, MVVM, UML, Agile

Full-Stack Developer — ICAR- SA

Sousse, Tunisia | Jan 2022 – Feb 2023

- Built a stock management web application (Spring Boot, Angular, MySQL, REST API) with a modular MVVM architecture to streamline inventory operations.
- Developed responsive interfaces with HTML, CSS, Bootstrap, Sass and JavaScript; designed and deployed the company's presentational website.

PROJECTS

Smart Health Platform — Microservices Healthcare System

Team project | *GitHub: smarthealth-platform*

- Developed, as a team, a healthcare management platform split into 9 gRPC microservices (patients, doctors, appointments, medical records, payments, inventory, laboratory, notifications, alerts).
- Exposed the services through a Node.js API Gateway offering REST and GraphQL (Apollo Server), with event-driven communication over Apache Kafka (10 business topics).
- React, Tailwind CSS and Apollo Client frontend; the whole stack is containerized with Docker Compose.

Technologies: Node.js, gRPC, GraphQL, Apache Kafka, React, Tailwind CSS, SQLite, RxDB, Docker

ENISO-LEARNING — Educational Web Platform

2025 | *GitHub: frontend · backend*

- Developed a web platform for ENISO where teachers upload and manage course materials and students download resources and follow updates.
- Implemented real-time teacher-student chat with WebSocket and an AI chatbot assistant powered by the Google Gemini API.

Technologies: Angular, Angular Material, Spring Boot, MySQL, WebSocket, Google Gemini API

Intelligent Quadcopter Drone

2025 | *GitHub: smart-quadcopter-drone-esp32*

- Designed an autonomous quadcopter: ESP32 flight control, PID stabilization, GPS, telemetry, brushless motors.
- Implemented low-latency ESP-NOW control with joystick input and automatic return on signal loss.
- Built a C# flight monitor (YMFC-32) showing GPS, altitude, orientation, temperature and flight modes.

Technologies: ESP32, ESP-NOW, C++, C#, GPS module, brushless motors

Vuforia AR App for Street Plaques

2024 | *GitHub: Unity-Vuforia-Road-Sign-Recognition-AR*

- Developed an augmented reality app with Unity and Vuforia that recognizes street-sign plaques through the camera and plays a short educational video about road signs.

CERTIFICATES

freeCodeCamp Certifications | Sep 2021

Front-End Development Libraries; JavaScript Algorithms and Data Structures; Responsive Web Design

Google UX Design Professional Certificate (Coursera) | Oct 2023

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Java, Python, PHP, Kotlin, C, C++, C#

Frontend: Angular, React.js, HTML, CSS, Sass, Bootstrap, Tailwind CSS, Material UI, jQuery, DOM

Backend: Spring Boot, Laravel, Symfony, PHP, MERN stack (MongoDB, Express.js, React, Node.js), REST APIs, GraphQL

Architecture & Messaging: Microservices, API Gateway, gRPC, Apache Kafka, MVC, MVVM

Mobile & Desktop: Flutter, Java, Kotlin, JavaFX

Databases & BaaS: MySQL, PostgreSQL, Oracle, SQLite, MongoDB, Mongoose, Firebase, Supabase

DevOps & Cloud: Docker, CI/CD, GitHub Actions, Azure, Google Cloud Platform (GCP), Linux, Bash, Git

Tools & Practices: GitHub, Postman, Figma, IntelliJ IDEA, VS Code, MySQL Workbench, UML, Agile

Embedded, AR/VR & Vision: ESP32, Arduino, STM32, Raspberry Pi, Vuforia, Unity, Leap Motion, Blender, OpenCV

Soft Skills: Leadership, event organization, teamwork, time management, problem-solving, documentation

LANGUAGES

English (Intermediate) | French (Excellent) | Arabic (Native)

LEADERSHIP

Chair — IEEE ENISO Student Branch

- Lead and coordinate the branch's volunteer members and activities.
- Plan and coordinate technical events for students, including **COPx 2.0** (event page).